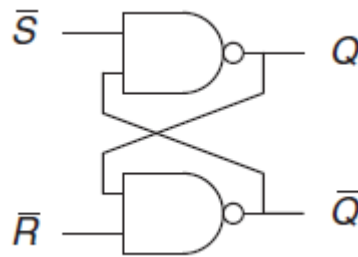


Lecture 03 Assignments

Sequential Logic Circuits

- 1) Describe the difference between *combinational logic circuits* and *sequential logic circuits*.
- 2) Is the circuit in the following figure combinational logic or sequential logic? Explain the relationship between inputs and outputs using a truth table. What would you call this circuit?



- 3) Design an S-R latch, and show its truth table. Show how to convert the S-R latch into a D latch. Indicate the advantage(s) of the D-latch over the S-R latch.
- 4) Design a D flip-flop using the D latch(s). Show how to construct a 4-bit **parallel-in-parallel-out** register from the D flip-flops. Describe the importance of registers in a computer system.
- 5) Describe how the designers eliminate the forbidden state of S-R latches and show the structure of the resulting latch with its truth table.
- 6) Design a J-K flip-flop from J-K latches. Show the circuit schematics of a **mod-13** ripple counter using the J-K flip-flops, and show the output sequence of the counter using a truth table.
- 7) Describe the term “toggling” in J-K latches. Show how to construct a J-K flip-flop out of J-K latches.
- 8) Draw the circuit schematics of a **mod-13** ripple counter using J-K flip-flops and show the output sequence of the counter using a truth table.
- 9) Draw the circuit schematics of a **mod-6** ripple counter using **toggling** flip-flops and show the output sequence of the counter using a truth table.
- 10) [4 Points] Draw the circuit schematics of a **mod-11** ripple counter using the J-K flip-flops and show the output sequence of the counter using a truth table.